In[ ]:=
```
n1 = 3;
g1 = 4;
n2 = 2;
g2 = 2;
n3 = 2;
g3 = 2;


(*Consistencies solution for the choice of pentaquark *)

γ12bar = 0.45566826601850596; (* relationship between light and anti-charm quark *)
β12 = 0.2204674444696147; (* relationship between light and charm quark *)
β22bar = 0.8237105803142551; (* relationship between charm and anti-charm quark *)

(*Defining Constants*)

xx = 0.346;
(*This is the value of αs which is αs=g^2/(ħc),
used from the mesonic fit as done in meson calculations.*)
alfas = xx;
(*aa=.178;*)
zz = 306.;(*This is the mass of light quark like up, down*)
aa = 0.197; (*This is the ratio of light mass to the heavy mass of quarks *)
yy = 107.6;
B0 = yy; (*Bag Constant *)
a1 = 1 / zz; (*This is α=ħ/(m1 c), where m1 is the mass of light quark. here,
reduced planck's constant and c both are set to be 1.*)
P = N[Pi, 7];
alfa1 = 1.;(* Ratio of mass of light quarks, BA, 13 September *)
alfa2 = aa; (* Ratio of mass of light quark to the heavy quark,
BA, 13 September *)

η = (n1 * g1 + n3 * g3 + n2 * g2) / 5;
theta = -6 η / (5 (5 η - 1)); (*Just a multiplication factor for a particular value of eta *)
Q1 = -0.6947368421052631`; (*Q1 is the constant in the differential equation  d^2 f1/dx^2 =
```

Q1 $\frac{f1^{\frac{3}{2}}}{\sqrt{x}}$ for the outer region*)

```
Q3 = -1.577809540840739;
(*Q3 is the constant in the differential equation  d^2 f3/dx^2 =
 Q3 f3^(3/2)/Sqrt[x] in the middle region. This is obtained by solving two
  consistency conditions along with other constant γ12bar. *)
Q2 = -3.7583491797205992;
(*Qbar is the constant in the differential equation  d^2 f3/dx^2 =
 Q2 f3^(3/2)/Sqrt[x] in the inner region. This is obtained by solving three
    consistency conditions along with other constants β13 & β23. *)
g = alfas;(*Why  g = alfas in this case ?*)
(* I think there should be g = /sqrt{alfas} in order to satisfy the multiplication
  facor in the expreesion of kinetic energy i.e. 'c', BA, 13 September *)
R = 3 a/(8 * alfas) * (3 π η/2)^(2/3);(*Page number 54, it is inverse in the copy though *)
c = 12/(5 π) 4/3 g^2 - 4/3 alfas/a (3 π η/2)^(1/3);
(*Multiplication factor in the expression of kinetic energy *)
d = 4/3 g^2 - 4/3 alfas/a (3 π η/2)^(1/3) 18/(25 π) η/(η - 1/5); (* 18g^2/(5(5η -1 )).16 π^2.(3a/8alfas). R^2/(6π^2)^2 ,
The value of R is from page number 54 which is written inverse here,
and (3a/8alfas) is inverse in copy, BA, 13 September *)
(*Multiplication factor in the expression of Potential energy *)


(*Let us define the three functions for three regions *)

f3[x3_?NumericQ, x2_?NumericQ, u2_?NumericQ , s2_?NumericQ] :=
  Abs[x3 * y3'[x3] - y3[x3] - n3 * Q3/(3 η) - β22bar * n2 * Q2/(3 η) + β22bar^(3/2) * n2 * Q3/(3 η)]
```

```
(*Appendix from Kinetic energy*) /.
 First@NDSolve[{y3''[x] == (Q3 * Abs[y3[x]^(3/2)] / x^(1/2)), y3'[x2] == β22bar * s2 ,
    y3[x2] == β22bar * u2 (* It shows how the function f3
     (x3) changes on other regions*)}, y3, {x, x2, x3}];
(*u2 is the function f_2(x) *)
(* s2 is the first derivative of function f_2(x) *)
(*This is the function f3 defined on the
  middle region. It has the differential equation  d^2 f3/dx^2 =
 Q3 f3^(3/2)/Sqrt[x] in for x3<x<x2 which is middle region. On this region, f1(x)=γ12 f2(x).  *)
f1[x1_?NumericQ, x3_?NumericQ, s3_?NumericQ, u3_?NumericQ] :=
  Abs[x1 * y1'[x1] - y1[x1] - n1 * Q1/(3 η) - γ12bar * n3 * Q3/(3 η) +
    γ12bar^(3/2) * n3 * Q1/(3 η) - (γ12bar^(3/2) * β22bar^(3/2) - β12^(3/2)) * n2 * Q1/(3 η) +
    γ12bar * β22bar^(3/2) * n2 * Q3/(3 η) - γ12bar * β22bar * n2 * Q2/(3 η)]
   (*Appendix from Kinetic energy*) /. First@NDSolve[{y1''[x] == (Q1 * Abs[y1[x]^(3/2)] / x^(1/2)),
     y1'[x3] == γ12bar * s3, (* s3 is the first derivative of function f_2bar(x) *)
     (* It means at the boundary x3 the function df1 changes to other function *)
     y1[x3] == γ12bar * u3 (* u3 is the function f_2bar(x) *)
     (*It means at the boundary x3 the function f1 changes to other function*)
    }, (* It means at the boundary x3 the function f1 changes to other function *)
    y1, {x, x3, x1}]; (*This is the function defined in outer region for x2<
  x<x1 so called f1. It has differential equation  d^2 f1/dx^2 =
 Q1 f1^(3/2)/Sqrt[x] for the outer region which is solved by using boundary
    condition of (d/dx) f1(x)|x2 = γ12bar (d/dx) f2(x)|x2 and f1(x3)=γ12 f2(x3)  *)
```

```
f2[xmax_?NumericQ, m2_?NumericQ] :=
  Module[{xx = xmax},
   solution2 = NDSolve[
     {y2''[x] == (Q2 * Abs[y2[x]^3/2]) / x^1/2, y2'[1.*^-9] == m2, y2[1.*^-9] == 0.}, y2, {x, 1.*^-9, xx}];
   m = NIntegrate[Sqrt[x] * Evaluate[y2[x] /. solution2]^(3 / 2),
     {x, 1.*^-9, xx}, Method → {Automatic, "SymbolicProcessing" → 0}];
   z = m[[1]];
   z = z - n2 / (3. * η);
   (*z=z-n/(A)*);
   Return[Abs[z]]; Remove["NDSolve`*$*$1"]];
(*This is the function f3 defined on the inner region. It has the differential
  equation  d^2 f3/dx^2 = Q3 f3^(3/2)/Sqrt[x]  for 0<x<x_3 inner region. On this region,
f_1(x)=β_13 f_3(x) and f_2(x)= β_23 f_3(x).    *)

(*Start the program*)

m2 = 25;
For[j = 1., j ≤ 1., j++, B = (B0)^4;
  a = a1; (* It must be to satisfy the multiplication
   factor used in kinetic energy i.e. c, BA, 13 September *)
  (*Now for the first pentaquark,
  eta=1 which in total contains 5 quarks which can be written by *)

 mqn = If[j > 1, m2 + 2, m2]; (*case2new*)(*mqn=If[j>1,m2+4,m2];*)
  m2 = mqn;
  topi = 0.01;
  m2B = m2;
  Energy = 10^9;
  Energy = 10^9;
  Energy2plus = 10^9;
```

```
H = 10^9 + 1;
SN = 0;
nloop = 1;
For[n = 1, n ≤ 3, n++, While[Energy ≤ H, F = H;
  H = Energy;
  HX = Energy6bare;
  HY = Energy2plus; m2 = m2 + SN * 10^(-n);
  x2 = 2.*^-8; (*Print[m2];*)(*was 0 .5 1.66*)
  FindMinimum[f2[x2, m2], {x2, topi}];
  (*remove this function and investigate if it is actually required BS 4/3/2021*)
  x2 = x2 /. Last[FindMinimum[f2[x2, m2], {x2, topi}]];
  solution2 = NDSolve[{y2''[x] == (Q2 * Abs[y2[x]^(3/2)]) / x^(1/2),
     y2'[1.*^-9] == m2, y2[1.*^-9] == 0.}, y2, {x, 1.*^-9, x2}];
  t1 = Evaluate[y2[x2] /. solution2];
  t4 = Evaluate[y2'[x2] /. solution2];
  u2 = t1[[1]]; (* u2 is the function f_2(x) *)
  s2 = t4[[1]]; (* s2 is the first derivative of function f_2(x) *)
  If[u2 < 0., u2 = 0.];


  x3 = x2 + 0.1;
  FindMinimum[{f3[x3, x2, u2, s2]}, {x3, x2}];
  x3 = x3 /. Last[FindMinimum[{f3[x3, x2, u2, s2]}, {x3, x2}]];
  solution3 = NDSolve[{y3''[x] == (Q3 * Abs[y3[x]^(3/2)]) / x^(1/2),
     y3'[x2] == β22bar * s2 , y3[x2] == β22bar * u2}, y3, {x, x2, x3}];
  t2 = Evaluate[y3[x3] /. solution3];
  t5 = Evaluate[y3'[x3] /. solution3];
  u3 = t2[[1]]; (* u3 is the function f_2bar(x) *)
  s3 = t5[[1]]; (* s3 is the first derivative of function f_2bar(x) *)
  If[u3 < 0., u3 = 0.];

  x1 = x3 + 0.1;
  FindMinimum[{f1[x1, x3, s3, u3]}, {x1, x3}];
  x1 = x1 /. Last[FindMinimum[{f1[x1, x3, s3, u3]}, {x1, x3}]];
```

$$\text{solution1} = \text{NDSolve}\left[\left\{y1''[x] == \left(Q1 * \frac{\text{Abs}\left[y1[x]^{\frac{3}{2}}\right]}{x^{\frac{1}{2}}}\right),\right.\right.$$

$$\left.\left. y1'[x3] == \gamma 12\text{bar} * s3 \, , \, y1[x3] == \gamma 12\text{bar} * u3 \right\}, y1, \{x, x3, x1\}\right];$$

```
t3 = Evaluate[y1[x1] /. solution1];
t6 = Evaluate[y1'[x1] /. solution1];
u1 = t3[[1]]; (*u1 is the function f_1(x) *)
s1 = t6[[1]]; (*s1 is the first derivative of function f_1(x) *)
If[u1 < 0., u1 = 0.];


(*Integrations *)(* These integrations change values but outer and outer 1
 doesnot because they depend on x and derivates of x (BA, 11 September)*)
```

$$\text{normmx2} = x2\, s2 - u2 ;$$

$$\text{normx2} = \frac{n2\, Q2}{3\, \eta} ;$$

```
(*normx2barr and normx2bar should be equal *)
```

$$\text{normmx3} = x3\, s3 - u3 ;$$

$$\text{normx3} = \frac{n3\, Q3}{3\, \eta} + \beta 22\text{bar} \frac{n2\, Q2}{3\, \eta} - (\beta 22\text{bar})^{\frac{3}{2}} \frac{n2\, Q3}{3\, \eta};$$

```
(*normmx2 and normx2 should be equal *)
```

$$\text{normmx1} = x1\, s1 - u1 ;$$

$$\text{normx1} = \frac{n1\, Q1}{3\, \eta} + \gamma 12\text{bar} \frac{n3\, Q3}{3\, \eta} - (\gamma 12\text{bar})^{\frac{3}{2}} \frac{n3\, Q1}{3\, \eta} +$$

$$\left((\gamma 12\text{bar})^{\frac{3}{2}} (\beta 22\text{bar})^{\frac{3}{2}} - (\beta 12)^{\frac{3}{2}}\right) \frac{n2\, Q1}{3\, \eta} - \gamma 12\text{bar} (\beta 22\text{bar})^{\frac{3}{2}} \frac{n2\, Q3}{3\, \eta} + \gamma 12\text{bar}\, \beta 22\text{bar} \frac{n2\, Q2}{3\, \eta} ;$$

```
(*normmx1 and normx1 should be equal *)


h1 = NIntegrate[Evaluate[y2[x] /. solution2]^2.5 / (x^0.5),
  {x, 0, x2}, Method → {Automatic, "SymbolicProcessing" → 0}];
h2 = NIntegrate[Evaluate[y3[x] /. solution3]^2.5 / (x^0.5), {x, x2, x3},
  Method → {Automatic, "SymbolicProcessing" → 0}];
h3 = NIntegrate[Evaluate[y1[x] /. solution1]^2.5 / (x^0.5), {x, x3, x1},
  Method → {Automatic, "SymbolicProcessing" → 0}];
h4 = h1[[1]];
h5 = h2[[1]];
```

```
h6 = h3[[1]] ;
AB = 4/7 Sqrt[x2] (u2)^(5/2) - 5 n2/(21 η) s2 ;
CD = 4/7 Sqrt[x3] (u3)^(5/2) - 4/7 (β22bar)^(5/2) Sqrt[x2] (u2)^(5/2) -
   5/(7 Q3) s3 (n3 Q3/(3 η) + β22bar n2 Q2/(3 η) - (β22bar)^(3/2) n2 Q3/(3 η)) + 5/(7 Q3) (β22bar)^2 n2 Q2/(3 η) s2 ;
CCD = 4/7 Sqrt[x3] (u3)^(5/2) - 4/7 Sqrt[x2] (β22bar)^(5/2) (u2)^(5/2) - 5/(7 Q3) s3 normmx3 +
   5/(7 Q3) (β22bar)^2 normmx2 s2 ;

EF = 4/7 Sqrt[x1] (u1)^(5/2) - 4/7 (γ12bar)^(5/2) Sqrt[x3] (u3)^(5/2) -
   5/(7 Q1) s1 (n1 Q1/(3 η) + γ12bar n3 Q3/(3 η) - (γ12bar)^(3/2) n3 Q1/(3 η) + n2 Q1/(3 η) ((γ12bar)^(3/2) (β22bar)^(3/2) - (β12)^(3/2)) -
      (γ12bar) (β22bar)^(3/2) n2 Q3/(3 η) + (γ12bar) (β22bar) n2 Q2/(3 η)) +
   5/(7 Q1) (γ12bar)^2 s3 (n3 Q3/(3 η) + β22bar n2 Q2/(3 η) - (β22bar)^(3/2) n2 Q3/(3 η)) ;
(* h4 and AB should be equal *)
(* h5 and CD should be equal *)
(* h6 and EF should be equal *)

K222 = n2/(3 η Q2) s2 - 1/Q2 h4 ;
K22 = 4 n2/(7 η Q2) s2 - 4/(7 Q2) Sqrt[x2] (u2)^(5/2) ;
(*K2bar2barr and K2bar2bar should be equal *)
K32 = - 4/(7 Q2) β22bar^1.5 Sqrt[x2] (u2)^(5/2) +
```

$$\frac{n2}{3\,\eta}\left(\frac{\beta22bar^{1.5}}{Q2}-\frac{\beta22bar}{Q3}+\frac{5*\beta22bar^{1.5}}{7\,Q2}\right)s2+\frac{n2}{3\,\eta\,Q3}s3\,;$$

$$K332=\frac{n2\,(\beta22bar)^{\frac{3}{2}}}{3\,\eta\,Q2}s2-\frac{n2\,\beta22bar}{3\,\eta\,Q3}s2+\frac{n2}{3\,\eta\,Q3}s3-\frac{(\beta22bar)^{\frac{3}{2}}}{Q2}h4;$$

(*K2bar2 and K2barr2 should be equal *)

$$K21=-\frac{4}{7\,Q2}\beta12^{1.5}\sqrt{x2}\;(u2)^{\frac{5}{2}}+\frac{n2}{3\,\eta}\left(\frac{5}{7\,Q2}\beta12^{1.5}+\frac{\beta12^{1.5}}{Q2}-\frac{\gamma12bar^{1.5}*\beta22bar}{Q3}\right)s2+$$

$$\frac{n2}{3\,\eta}\left(\frac{\gamma12bar^{1.5}}{Q3}-\frac{\gamma12bar}{Q1}\right)s3+\frac{n2}{3\,\eta\,Q1}s1\,;$$

$$K221=\frac{n2}{3\,\eta\,Q2}(\beta12)^{\frac{3}{2}}s2+\frac{n2\,(\gamma12bar)^{\frac{3}{2}}}{3\,\eta\,Q3}s3-$$

$$\frac{n2}{3\,\eta\,Q3}(\gamma12bar)^{\frac{3}{2}}\beta22bar\,s2+\frac{n2}{3\,\eta\,Q1}s1-\frac{n2}{3\,\eta\,Q1}\gamma12bar\,s3-\frac{(\beta12)^{\frac{3}{2}}}{Q2}h4\,;$$

(*K2bar1 and K2barr1 should be equal *)

$$K31=\frac{4}{7}\beta22bar^{1.5}\left(\frac{\gamma12bar^{1.5}*\beta22bar}{Q3}-\frac{\beta12^{1.5}}{Q2}\right)\sqrt{x2}\;(u2)^{\frac{5}{2}}-\frac{4}{7\,Q3}\gamma12bar^{1.5}\sqrt{x3}\;(u3)^{\frac{5}{2}}+$$

$$\frac{n2}{3\,\eta}\beta22bar\left(\frac{5}{7\,Q2}\beta22bar^{\frac{1}{2}}*\beta12^{1.5}-\frac{\gamma12bar^{1.5}*\beta22bar^{1.5}}{Q3}-\frac{5\,Q2}{7\,Q3^2}\gamma12bar^{1.5}*\right.$$

$$\left.\beta22bar+\frac{Q2}{Q3^2}\gamma12bar^{1.5}*\beta22bar+\frac{1}{Q2}\beta12^{1.5}*\beta22bar^{\frac{1}{2}}-\frac{\beta12^{1.5}}{Q3}\right)s2+$$

$$\frac{1}{3\,\eta\,Q3}\left(n3*\gamma12bar^{1.5}-\frac{n3\,Q3}{Q1}\gamma12bar+n2*\beta12^{1.5}-\frac{n2\,Q2}{Q3}\gamma12bar^{1.5}*\beta22bar+\right.$$

$$\left.\frac{5}{7}\gamma12bar^{1.5}\left(n3+\frac{n2\,Q2}{Q3}\beta22bar-n2*\beta22bar^{1.5}\right)\right)s3+\frac{n3}{3\,\eta\,Q1}s1\,;$$

$$(*K122 = \frac{n2}{3\,\eta}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}\left[\frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] - \frac{(\beta 12)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}}{Q2}h4 - \frac{(\gamma 12bar)^{\frac{3}{2}}}{Q3}h5$$

$$- \frac{n2\,Q2}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\left[\frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] + \frac{n3}{3\,\eta\,Q1}s1 - \frac{n3}{3\,\eta\,Q1}\gamma 12bar\,s3$$

$$+ (\beta 12)^{\frac{3}{2}}\frac{n2}{3\,\eta}\left[\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2}s2 + \frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] +$$

$$\frac{n3}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}s3\ ;\ *)$$

$$K331 = \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}\beta 22bar\,s2 - \frac{(\beta 12)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}}{Q2}h4 - \frac{(\gamma 12bar)^{\frac{3}{2}}}{Q3}h5 -$$

$$\frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s3 + \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^2\,s2 + \frac{n2}{3\,\eta}(\beta 22bar)^{\frac{3}{2}}$$

$$\left(\frac{\beta 12^{\frac{3}{2}}}{Q2} - \frac{(\gamma 12bar)^{\frac{3}{2}}\beta 22bar}{Q3}\right)s2 + \frac{n3}{3\,\eta}\gamma 12bar\left(\frac{\gamma 12bar^{\frac{1}{2}}}{Q3} - \frac{1}{Q1}\right)s3 + \frac{n3}{3\,\eta\,Q1}s1\ ;$$

(*K12 and K122 should be equal*)

$$K33 = \frac{4}{7}\beta 22bar^2\left(\frac{\beta 22bar^{\frac{1}{2}}}{Q3} - \frac{\beta 22bar}{Q2}\right)\sqrt{x2}\,(u2)^{\frac{5}{2}} -$$

$$\frac{4}{7\,Q3}\sqrt{x3}\,(u3)^{\frac{5}{2}} + \frac{n2}{3\,\eta}\beta 22bar^2\left(\beta 22bar\frac{12}{7\,Q2} + \frac{2}{7}\frac{Q2}{Q3^2} - \frac{2*\beta 22bar^{\frac{1}{2}}}{Q3}\right)s2 +$$

$$\left(\frac{12\,n3}{21\,\eta\,Q3} + \beta 22bar\frac{n2}{3\,\eta\,Q3}\left(\beta 22bar^{\frac{1}{2}} - \frac{2}{7}\frac{Q2}{Q3} - \frac{5}{7}\beta 22bar^{\frac{1}{2}}\right)\right)s3\ ;$$

$$K333 = \frac{n2\,Q2}{3\,\eta\,Q3}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2} - \frac{\beta 22bar}{Q3}\right)s3 - \frac{n2\,Q2\,\beta 22bar}{3\,\eta\,Q3}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2} - \frac{\beta 22bar}{Q3}\right)s2 -$$

$$\frac{(\beta 22bar)^3}{Q2}h4 - \frac{1}{Q3}h5 + (\beta 22bar)^3\frac{n2}{3\,\eta\,Q2}s2 - (\beta 22bar)^{\frac{5}{2}}\frac{n2}{3\,\eta\,Q3}s2 + \frac{n3}{3\,\eta\,Q3}s3\ ;$$

(*K22 and K222 should be equal *)

(*THese Y and Z are to check the value of integral of power $\frac{5}{2}$ in the outer region. Which proved to be false yet. I need to work on it extensively.*)

$$K11 = \frac{4}{7}\left(\frac{\beta22bar^{\frac{5}{2}} * \gamma12bar^3}{Q3} - \frac{\beta12^3}{Q2}\right)\sqrt{x2}\ (u2)^{\frac{5}{2}} + \frac{4}{7}\left(\frac{\gamma12bar^{\frac{5}{2}}}{Q1} - \frac{\gamma12bar^3}{Q3}\right)\sqrt{x3}\ (u3)^{\frac{5}{2}} - \frac{4}{7\,Q1}\sqrt{x1}\ (u1)^{\frac{5}{2}} + \left(\frac{n2}{3\,\eta}\left(\frac{\beta12^3}{Q2} - \frac{\beta12^{\frac{3}{2}} * \gamma12bar^{\frac{3}{2}} * \beta22bar}{Q3} + \frac{Q2}{Q3^2}\gamma12bar^3 * \beta22bar^2 - \frac{\beta12^{1.5} * \gamma12bar^{1.5} * \beta22bar}{Q3}\right) + \frac{5\,n2}{21\,\eta}\left(\frac{\beta12^3}{Q2} - \frac{Q2}{Q3^2}\beta22bar^2 * \gamma12bar^3\right)\right)s2 + \left(\frac{n2}{3\,\eta}\left(\frac{\beta12^{1.5} * \gamma12bar^{1.5}}{Q3} - \frac{\beta12^{1.5} * \gamma12bar}{Q1} - \frac{\beta22bar^{1.5} * \gamma12bar^3}{Q3} + \frac{\beta22bar^{1.5} * \gamma12bar^{\frac{5}{2}}}{Q1} + \frac{\beta12^{1.5} * \gamma12bar^{1.5}}{Q3} - \frac{Q2}{Q3^2}\gamma12bar^3 * \beta22bar - \frac{\beta12^{1.5} * \gamma12bar}{Q1} + \frac{\gamma12bar^{\frac{5}{2}} * \beta22bar^{1.5}}{Q1} + \frac{Q2}{Q1^2}\gamma12bar^2 * \beta22bar - \frac{Q3}{Q1^2}\beta22bar^{1.5} * \gamma12bar^2\right) + \frac{n3}{3\,\eta}\left(\frac{\gamma12bar^3}{Q3} - \frac{2 * \gamma12bar^{\frac{5}{2}}}{Q1} + \frac{Q3}{Q1^2}\gamma12bar^2\right) + \frac{5}{21\,\eta}\left(\frac{n3}{Q3}\gamma12bar^3 + \frac{n2\ Q2}{Q3^2}\gamma12bar^3 * \beta22bar - \frac{n2}{Q3}\beta22bar^{1.5} * \gamma12bar^3 - \frac{n3\ Q3}{Q1^2}\gamma12bar^2 - \frac{n2\ Q2}{Q1^2}\gamma12bar^2 * \beta22bar + \frac{n2\ Q3}{Q1^2}\gamma12bar^2 * \beta22bar^{1.5}\right)\right) * s3 + \left(\frac{n1}{3\,\eta\,Q1} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\gamma12bar + \frac{n3}{3\,\eta\,Q1}\gamma12bar^{1.5} + \frac{n2}{3\,\eta\ Q1}$$

$$\left(\beta 12^{1.5} - \gamma 12bar^{1.5} * \beta 22bar^{1.5} - \frac{Q2}{Q1}\beta 22bar * \gamma 12bar + \frac{Q3}{Q1}\beta 22bar^{1.5} * \gamma 12bar\right) + \frac{5}{21\,\eta}\left(\frac{n1}{Q1} + \frac{n3\ Q3}{Q1^2}\gamma 12bar - \frac{n3}{Q1}\gamma 12bar^{1.5} + \frac{n2}{Q1}\gamma 12bar^{1.5} * \beta 22bar^{1.5} - \frac{n2}{Q1}\beta 12^{1.5} - \frac{n2\ Q3}{Q1^2}\beta 22bar^{1.5} * \gamma 12bar + \frac{n2\ Q2}{Q1^2}\gamma 12bar\ \beta 22bar\right)\Bigg) * s1;$$

$$K111 = \frac{-(\beta 12)^3}{Q2}h4 - \frac{(\gamma 12bar)^3}{Q3}h5 - \frac{1}{Q1}h6 + \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^3\,\beta 22bar\,s3 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s2 + \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^3(\beta 22bar)^2 s2 + \left(\frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}} + \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{3}{2}} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\gamma 12bar - \frac{n2}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}} - \frac{n2\,Q2}{3\,\eta\,Q1^2}\beta 22bar\,\gamma 12bar + \frac{n2\,Q3}{3\,\eta\,Q1^2}(\beta 22bar)^{\frac{3}{2}}\gamma 12bar\right)s1 - \left(\frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}}\gamma 12bar + \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}} - \frac{n3\,Q3}{3\,\eta\,Q1^2}(\gamma 12bar)^2 - \frac{n2}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}}(\beta 22bar)^{\frac{3}{2}} - \frac{n2\,Q2}{3\,\eta\,Q1^2}(\gamma 12bar)^2\,\beta 22bar + \frac{n2\,Q3}{3\,\eta\,Q1^2}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^2\right)s3 + \frac{n2}{3\,\eta}\frac{(\beta 12)^3}{Q2}s2 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s2 + \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}}\gamma 12bar\,s3 + \frac{n3}{3\,\eta\,Q3}(\gamma 12bar)^3 s3 - \frac{n2}{3\,\eta\,Q3}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^3 s3 - \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}}s3 + \frac{n2}{3\,\eta\,Q1}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^{\frac{5}{2}}s3 + \frac{n1}{3\,\eta\,Q1}s1;$$

(*K11 and K111 should be equal *)

$$T1 = c\left(\frac{4}{7}\left(alfa1 * g1\left(\beta 12^{\frac{5}{2}} - \gamma 12bar^{\frac{5}{2}} * \beta 22bar^{\frac{5}{2}}\right) + alfa2 * g2\right)\sqrt{x2}\ (u2)^{\frac{5}{2}} +$$

$$\frac{4}{7}\,\text{alfa2}*\text{g3}\sqrt{\text{x3}}\,(\text{u3})^{\frac{5}{2}}+\frac{4}{7}\,\text{alfa1}*\text{g1}\sqrt{\text{x1}}\,(\text{u1})^{\frac{5}{2}}+$$

$$\frac{5\,\text{n2}}{21\,\eta}\left(\beta 22\text{bar}^{2}\,\frac{\text{Q2}}{\text{Q3}}\left(\text{alfa1}*\text{g1}*\gamma 12\text{bar}^{\frac{5}{2}}+\text{alfa2}*\text{g3}\right)-\right.$$

$$\left.\text{alfa1}*\text{g1}*\beta 12^{\frac{5}{2}}-\text{alfa2}*\text{g3}*\beta 22\text{bar}^{\frac{5}{2}}-\text{alfa2}*\text{g2}\right)\text{s2}+$$

$$\frac{5}{21\,\eta}\left(\frac{\gamma 12\text{bar}^{2}*\text{alfa1}*\text{g1}}{\text{Q1}}\left(\text{n3}*\text{Q3}+\beta 22\text{bar}*\text{n2}*\text{Q2}-\beta 22\text{bar}^{1.5}*\text{n2}*\text{Q3}\right)-\right.$$

$$\left.\left(\text{alfa1}*\text{g1}*\gamma 12\text{bar}^{\frac{5}{2}}+\text{alfa2}*\text{g3}\right)\left(\text{n3}+\beta 22\text{bar}\,\frac{\text{n2}*\text{Q2}}{\text{Q3}}-\beta 22\text{bar}^{1.5}*\text{n2}\right)\right)\text{s3}-$$

$$\frac{5\,\text{alfa1}\,\text{g1}}{21\,\eta}\left(\text{n1}+\gamma 12\text{bar}\,\frac{\text{n3}\,\text{Q3}}{\text{Q1}}+\gamma 12\text{bar}^{1.5}\left(\beta 22\text{bar}^{1.5}*\text{n2}-\text{n3}\right)-\right.$$

$$\left.\beta 12^{1.5}*\text{n2}-\gamma 12\text{bar}*\beta 22\text{bar}^{1.5}\,\frac{\text{n2}\,\text{Q3}}{\text{Q1}}+\gamma 12\text{bar}\,\beta 22\text{bar}\,\frac{\text{n2}\,\text{Q2}}{\text{Q1}}\right)\text{s1}\Bigg);$$

$$\text{V1}=-\text{d}\left(\frac{(\text{g1}\,\text{n1}-1)}{\text{n1}}\,\text{g1}\,\text{K11}+\frac{(\text{g3}\,\text{n3}-1)}{\text{n3}}\,\text{g3}\,\text{K33}+\right.$$

$$\left.\frac{(\text{g2}\,\text{n2}-1)}{\text{n2}}\,\text{g2}\,\text{K22}+2\,\text{g1}\,\text{g3}\,\text{K31}+\frac{4}{3}\,\text{g2}\,\text{g1}\,\text{K21}+\frac{4}{3}\,\text{g2}\,\text{g3}\,\text{K32}\right);$$

```
(*These T1 and V1 are obtained
 by integrating manually which is being verified.*)
Energy6bare = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B + n1 * g1 * 306 + n3 * g3 * 1553 + n2 * g2 * 1553;
Energy = Energy6bare;
Volumeenergy = 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Energywithoutmass = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Print["R=  ", R];
Print["normx2 (computationally)=  ", normmx2];
Print["normx2 (by hand)=  ", normx2];
Print["normx3 (computationally)=  ", normmx3];
Print["normx3 (by hand)=  ", normx3];
Print["normx1 (computationally)=  ", normmx1];
Print["normx1 (by hand)=  ", normx1];
Print["0 to x2 (h4 computationally)=  ", h4];
Print["0 to x2 (h4 by hand)=  ", AB];
```

```
 Print["x2 to x3 (h5 computationally)=  ", h5];
 Print["x2 to x3 (h5 by hand)=  ", CD];
 Print["x2 to x3 (h5 by handdddd)=  ", CCD];
 Print["x3 to x1 (h6 computationally)=  ", h6];
 Print["x3 to x1 (h6 by hand)=  ", EF];
 Print["K22  (computationally) = ", K222];
 Print["K22  (by hand) = ", K22];
 Print["K32  (computationally) = ", K332];
 Print["K32  (by hand) = ", K32];
 Print["K21  (computationally) = ", K221];
 Print["K21  (by hand) =", K21];
 Print["K31  (computationally) = ", K331];
 Print["K31  (by hand) = ", K31];
 Print["K33  (computationally) = ", K333];
 Print["K33  (by hand) = ", K33];
 Print["K11  (computationally) = ", K111];
 Print["K11  (by hand) = ", K11];
 Print["x2= ", x2];
 Print["x3= ", x3];
 Print["x1=  ", x1];
  Print["T by hand=", T1];
 Print["V by hand=", V1];
 Print["m2=", m2];
 Print["Energy without mass = ", Energywithoutmass];
 Print["Energy=  ", Energy];
 If[n == 1, If[nloop == 1, nloop = nloop + 1; SN = 1. * 10^(-1);
   Energy11 = Energy, If[nloop == 2, nloop = nloop + 1;
    H = 10^9 + 1;
    m2 = m2B;
    If[Energy - Energy11 < 0, SN = 1., SN = -1.]]]];
If[n < 3, m2 = m2 - SN * 10^(-n); Energy = F; H = F];
If[n == 3, Energy6bare = HX;
 Print["x1=", x1];
 Print["x3=", x3];
 Print["x2=", x2];
 Print["Kinetic Energy=", T1];
 Print["Potential Energy=", V1];
 Print["Volume Energy=", Volumeenergy];
]]];

(*Code for Plotting function *)
Plot1 = Plot[Evaluate[((y2[x] /.solution2)/ x)^(3 / 2)],
```

```
  {x, .00001, x2}, (*%PlotRange→{{0,0.65},{0,79.5}},*)
  (*Solution 1 = f3(x3), solution3 = f3(x3), solution3 = f1(x1) *)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17]
    , StyleForm["(f2(x)/x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot2 = Plot[Evaluate[((β22bar * y2[x] /.solution2) / x)^(3 / 2)], {x, .00001, x2},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f2 bar(x)/x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot3 = Plot[Evaluate[((y3[x] /.solution3) / x)^(3 / 2)], {x, x2, x3},
  (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f2 bar(x)/x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot4 = Plot[Evaluate[((β12 * y2[x] /.solution2) / x)^(3 / 2)], {x, .00001, x2},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot5 = Plot[Evaluate[((γ12bar * y3[x] /. solution3) / x)^(3 / 2)],
  {x, x2, x3}, (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot6 = Plot[Evaluate[((y1[x] /. solution1) / x)^(3 / 2)], {x, x3, x1},
  (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Show[{Plot2, Plot3}]
Show[{Plot4, Plot5, Plot6}]
R=  0.0250861775
normx2 (computationally)=  -0.626391453
normx2 (by hand)=  -0.62639153
normx3 (computationally)=  -0.582341804
normx3 (by hand)=  -0.582341835
normx1 (computationally)=  -0.418062719
normx1 (by hand)=  -0.418062719
0 to x2 (h4 computationally)=  3.96760981
0 to x2 (h4 by hand)=  3.96760977
```

x2 to x3 (h5 computationally)= 0.789888369

x2 to x3 (h5 by hand)= 0.789888861

x2 to x3 (h5 by handdddd)= 0.789888619

x3 to x1 (h6 computationally)= 1.75416399

x3 to x1 (h6 by hand)= 1.75416386

K22 (computationally) = 0.206343804

K22 (by hand) = 0.206343793

K32 (computationally) = 0.195115521

K32 (by hand) = 0.195115513

K21 (computationally) = 0.175179374

K21 (by hand) =0.175179373

K31 (computationally) = 0.173963129

K31 (by hand) = 0.173963224

K33 (computationally) = 0.186546122

K33 (by hand) = 0.186546427

K11 (computationally) = 0.22579783

K11 (by hand) = 0.225797673

x2= 0.162722275

x3= 0.180287016

x1= 0.327561917

T by hand=488.311275

V by hand=-147.319886

m2=25

Energy without mass = 652.536816

Energy= 16748.5368

R= 0.0250861775

normx2 (computationally)= -0.626391424

normx2 (by hand)= -0.62639153

normx3 (computationally)= -0.582341804

normx3 (by hand)= -0.582341835

normx1 (computationally)= -0.418062719

normx1 (by hand)= -0.418062719

0 to x2 (h4 computationally)= 3.96923687

0 to x2 (h4 by hand)= 3.96923672

x2 to x3 (h5 computationally)= 0.79022012

x2 to x3 (h5 by hand)= 0.790220781

```
x2 to x3 (h5 by handdddd)=  0.790220371
x3 to x1 (h6 computationally)=  1.75492865
x3 to x1 (h6 by hand)=  1.75492853
K22  (computationally) = 0.206385762
K22  (by hand) = 0.206385723
K32  (computationally) = 0.19515531
K32  (by hand) = 0.195155281
K21  (computationally) = 0.175216238
K21  (by hand) =0.175216234
K31  (computationally) = 0.173999803
K31  (by hand) = 0.173999929
K33  (computationally) = 0.186584401
K33  (by hand) = 0.186584799
K11  (computationally) = 0.225845953
K11  (by hand) = 0.225845809
x2= 0.162688893
x3= 0.180249816
x1=  0.327491123
T by hand=488.520173
V by hand=-147.350838
m2=25.01
Energy without mass = 652.512807
Energy=  16748.5128
R=  0.0250861775
normx2 (computationally)=  -0.626391473
normx2 (by hand)=  -0.62639153
normx3 (computationally)=  -0.582341804
normx3 (by hand)=  -0.582341835
normx1 (computationally)=  -0.418062715
normx1 (by hand)=  -0.418062719
0 to x2 (h4 computationally)=  3.98388258
0 to x2 (h4 by hand)=  3.98388262
x2 to x3 (h5 computationally)=  0.793203003
x2 to x3 (h5 by hand)=  0.793203376
x2 to x3 (h5 by handdddd)=  0.793203255
x3 to x1 (h6 computationally)=  1.76181108
```

```
x3 to x1 (h6 by hand)=  1.76181093
K22  (computationally) = 0.20676359
K22  (by hand) = 0.206763599
K32  (computationally) = 0.195513336
K32  (by hand) = 0.195513343
K21  (computationally) = 0.175547701
K21  (by hand) =0.175547702
K31  (computationally) = 0.174329045
K31  (by hand) = 0.174329119
K33  (computationally) = 0.186927068
K33  (by hand) = 0.18692731
K11  (computationally) = 0.226278374
K11  (by hand) = 0.226278175
x2= 0.162389401
x3= 0.179916005
x1=  0.326855973
T by hand=490.400241
V by hand=-147.628825
m2=25.1
Energy without mass = 652.306903
Energy=  16748.3069
R=  0.0250861775
normx2 (computationally)=  -0.626391473
normx2 (by hand)=  -0.62639153
normx3 (computationally)=  -0.582341804
normx3 (by hand)=  -0.582341835
normx1 (computationally)=  -0.418062717
normx1 (by hand)=  -0.418062719
0 to x2 (h4 computationally)=  4.00015601
0 to x2 (h4 by hand)=  4.00015605
x2 to x3 (h5 computationally)=  0.796518136
x2 to x3 (h5 by hand)=  0.79651851
x2 to x3 (h5 by handdddd)=  0.796518391
x3 to x1 (h6 computationally)=  1.76945943
x3 to x1 (h6 by hand)=  1.76945929
K22  (computationally) = 0.207182502
```

```
K22  (by hand) = 0.207182511
K32  (computationally) = 0.195910339
K32  (by hand) = 0.195910346
K21  (computationally) = 0.175915292
K21  (by hand) =0.175915293
K31  (computationally) = 0.174694264
K31  (by hand) = 0.174694338
K33  (computationally) = 0.187307355
K33  (by hand) = 0.187307598
K11  (computationally) = 0.226758048
K11  (by hand) = 0.226757868
x2= 0.162058544
x3= 0.179547257
x1=  0.326154483
T by hand=492.489491
V by hand=-147.937193
m2=25.2
Energy without mass = 652.099108
Energy=  16748.0991
R=  0.0250861775
normx2 (computationally)=  -0.626391432
normx2 (by hand)=  -0.62639153
normx3 (computationally)=  -0.582341804
normx3 (by hand)=  -0.582341835
normx1 (computationally)=  -0.418062707
normx1 (by hand)=  -0.418062719
0 to x2 (h4 computationally)=  4.01642997
0 to x2 (h4 by hand)=  4.01642985
x2 to x3 (h5 computationally)=  0.799833972
x2 to x3 (h5 by hand)=  0.799834593
x2 to x3 (h5 by handdddd)=  0.799834229
x3 to x1 (h6 computationally)=  1.77710883
x3 to x1 (h6 by hand)=  1.77710862
K22  (computationally) = 0.207600503
K22  (by hand) = 0.207600472
K32  (computationally) = 0.196306517
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```
K32  (by hand) = 0.196306494
K21  (computationally) = 0.176282137
K21  (by hand) =0.176282134
K31  (computationally) = 0.175058811
K31  (by hand) = 0.17505893
K33  (computationally) = 0.187687091
K33  (by hand) = 0.187687467
K11  (computationally) = 0.22723659
K11  (by hand) = 0.227236325
x2= 0.161729682
x3= 0.17918075
x1=  0.325457391
T by hand=494.578971
V by hand=-148.244913
m2=25.3
Energy without mass = 651.913114
Energy=  16747.9131
R=  0.0250861775
normx2 (computationally)=  -0.626391452
normx2 (by hand)=  -0.62639153
normx3 (computationally)=  -0.582341804
normx3 (by hand)=  -0.582341835
normx1 (computationally)=  -0.418062709
normx1 (by hand)=  -0.418062719
0 to x2 (h4 computationally)=  4.03270507
0 to x2 (h4 by hand)=  4.03270502
x2 to x3 (h5 computationally)=  0.803149503
x2 to x3 (h5 by hand)=  0.803150008
x2 to x3 (h5 by handdddd)=  0.803149762
x3 to x1 (h6 computationally)=  1.78475953
x3 to x1 (h6 by hand)=  1.78475933
K22  (computationally) = 0.208017795
K22  (by hand) = 0.208017783
K32  (computationally) = 0.196701967
K32  (by hand) = 0.196701959
K21  (computationally) = 0.176648266
```

```
K21  (by hand) =0.176648265
K31  (computationally) = 0.175422545
K31  (by hand) = 0.175422642
K33  (computationally) = 0.188065776
K33  (by hand) = 0.18806609
K11  (computationally) = 0.227714328
K11  (by hand) = 0.227714066
x2= 0.161402804
x3= 0.17881645
x1=  0.324764662
T by hand=496.668722
V by hand=-148.552034
m2=25.4
Energy without mass = 651.74864
Energy=  16747.7486
R=  0.0250861775
normx2 (computationally)=  -0.626391473
normx2 (by hand)=  -0.62639153
normx3 (computationally)=  -0.582341804
normx3 (by hand)=  -0.582341835
normx1 (computationally)=  -0.418062707
normx1 (by hand)=  -0.418062719
0 to x2 (h4 computationally)=  4.04898095
0 to x2 (h4 by hand)=  4.04898098
x2 to x3 (h5 computationally)=  0.806465315
x2 to x3 (h5 by hand)=  0.806465696
x2 to x3 (h5 by handdddd)=  0.806465576
x3 to x1 (h6 computationally)=  1.79241138
x3 to x1 (h6 by hand)=  1.79241117
K22  (computationally) = 0.208434268
K22  (by hand) = 0.208434277
K32  (computationally) = 0.197096641
K32  (by hand) = 0.197096647
K21  (computationally) = 0.177013668
K21  (by hand) =0.177013669
K31  (computationally) = 0.175785554
```

K31 (by hand) = 0.175785629

K33 (computationally) = 0.188443709

K33 (by hand) = 0.188443956

K11 (computationally) = 0.228191078

K11 (by hand) = 0.228190794

x2= 0.161077884

x3= 0.178454343

x1= 0.324076245

T by hand=498.758717

V by hand=-148.85853

m2=25.5

Energy without mass = 651.605363

Energy= 16747.6054

R= 0.0250861775

normx2 (computationally)= -0.62639139

normx2 (by hand)= -0.62639153

normx3 (computationally)= -0.582341804

normx3 (by hand)= -0.582341835

normx1 (computationally)= -0.41806271

normx1 (by hand)= -0.418062719

0 to x2 (h4 computationally)= 4.06525694

0 to x2 (h4 by hand)= 4.06525667

x2 to x3 (h5 computationally)= 0.809782456

x2 to x3 (h5 by hand)= 0.80978334

x2 to x3 (h5 by handdddd)= 0.809782719

x3 to x1 (h6 computationally)= 1.80006444

x3 to x1 (h6 by hand)= 1.80006424

K22 (computationally) = 0.208849723

K22 (by hand) = 0.20884965

K32 (computationally) = 0.197490442

K32 (by hand) = 0.197490387

K21 (computationally) = 0.177378344

K21 (by hand) =0.177378337

K31 (computationally) = 0.176148018

K31 (by hand) = 0.176148185

K33 (computationally) = 0.18882142

```
K33  (by hand) = 0.188821939
K11  (computationally) = 0.228666995
K11  (by hand) = 0.228666764
x2= 0.160754895
x3= 0.178094418
x1=  0.323392102
T by hand=500.849015
V by hand=-149.164536
m2=25.6
Energy without mass = 651.482939
Energy=  16747.4829
R=  0.0250861775
normx2 (computationally)=  -0.62639139
normx2 (by hand)=  -0.62639153
normx3 (computationally)=  -0.582341804
normx3 (by hand)=  -0.582341835
normx1 (computationally)=  -0.418062711
normx1 (by hand)=  -0.418062719
0 to x2 (h4 computationally)=  4.08153422
0 to x2 (h4 by hand)=  4.08153394
x2 to x3 (h5 computationally)=  0.813099063
x2 to x3 (h5 by hand)=  0.813099952
x2 to x3 (h5 by handdddd)=  0.813099328
x3 to x1 (h6 computationally)=  1.80771863
x3 to x1 (h6 by hand)=  1.80771844
K22  (computationally) = 0.209264526
K22  (by hand) = 0.209264452
K32  (computationally) = 0.197883552
K32  (by hand) = 0.197883497
K21  (computationally) = 0.177742309
K21  (by hand) =0.177742302
K31  (computationally) = 0.176509636
K31  (by hand) = 0.176509804
K33  (computationally) = 0.18919798
K33  (by hand) = 0.189198503
K11  (computationally) = 0.229141909
```

```
K11  (by hand) = 0.229141685
x2= 0.160433833
x3= 0.177736633
x1=  0.322712181
T by hand=502.939528
V by hand=-149.469859
m2=25.7
Energy without mass = 651.381153
Energy=  16747.3812
R=  0.0250861775
normx2 (computationally)=  -0.626391463
normx2 (by hand)=  -0.62639153
normx3 (computationally)=  -0.582341804
normx3 (by hand)=  -0.582341835
normx1 (computationally)=  -0.41806271
normx1 (by hand)=  -0.418062719
0 to x2 (h4 computationally)=  4.09781271
0 to x2 (h4 by hand)=  4.09781271
x2 to x3 (h5 computationally)=  0.816415216
x2 to x3 (h5 by hand)=  0.816415664
x2 to x3 (h5 by handdddd)=  0.816415483
x3 to x1 (h6 computationally)=  1.81537397
x3 to x1 (h6 by hand)=  1.81537377
K22  (computationally) = 0.209678666
K22  (by hand) = 0.209678665
K32  (computationally) = 0.198275966
K32  (by hand) = 0.198275966
K21  (computationally) = 0.178105568
K21  (by hand) =0.178105568
K31  (computationally) = 0.176870425
K31  (by hand) = 0.176870512
K33  (computationally) = 0.189573436
K33  (by hand) = 0.18957372
K11  (computationally) = 0.229615838
K11  (by hand) = 0.229615582
x2= 0.160114679
```

```
x3= 0.177380966
x1=  0.322036439
T by hand=505.030258
V by hand=-149.774513
m2=25.8
Energy without mass = 651.299713
Energy=  16747.2997
R=  0.0250861775
normx2 (computationally)=  -0.62639139
normx2 (by hand)=  -0.62639153
normx3 (computationally)=  -0.582341804
normx3 (by hand)=  -0.582341835
normx1 (computationally)=  -0.418062719
normx1 (by hand)=  -0.418062719
0 to x2 (h4 computationally)=  4.11409102
0 to x2 (h4 by hand)=  4.11409075
x2 to x3 (h5 computationally)=  0.819733137
x2 to x3 (h5 by hand)=  0.819734036
x2 to x3 (h5 by handdddd)=  0.819733406
x3 to x1 (h6 computationally)=  1.82303055
x3 to x1 (h6 by hand)=  1.82303041
K22  (computationally) = 0.210091716
K22  (by hand) = 0.210091642
K32  (computationally) = 0.198667481
K32  (by hand) = 0.198667425
K21  (computationally) = 0.178468112
K21  (by hand) =0.178468104
K31  (computationally) = 0.177230757
K31  (by hand) = 0.177230926
K33  (computationally) = 0.189948906
K33  (by hand) = 0.189949434
K11  (computationally) = 0.230089051
K11  (by hand) = 0.230088904
x2= 0.159797395
x3= 0.177027424
x1=  0.321364848
```

T by hand=507.12132

V by hand=−150.078753

m2=25.9

Energy without mass = 651.238239

Energy= 16747.2382

R= 0.0250861775

normx2 (computationally)= −0.626391431

normx2 (by hand)= −0.62639153

normx3 (computationally)= −0.582341804

normx3 (by hand)= −0.582341835

normx1 (computationally)= −0.418062711

normx1 (by hand)= −0.418062719

0 to x2 (h4 computationally)= 4.13037081

0 to x2 (h4 by hand)= 4.13037068

x2 to x3 (h5 computationally)= 0.82305019

x2 to x3 (h5 by hand)= 0.823050844

x2 to x3 (h5 by handdddd)= 0.823050461

x3 to x1 (h6 computationally)= 1.83068808

x3 to x1 (h6 by hand)= 1.83068788

K22 (computationally) = 0.210504193

K22 (by hand) = 0.210504159

K32 (computationally) = 0.199058348

K32 (by hand) = 0.199058322

K21 (computationally) = 0.17882995

K21 (by hand) =0.178829946

K31 (computationally) = 0.17759019

K31 (by hand) = 0.177590315

K33 (computationally) = 0.190323073

K33 (by hand) = 0.190323469

K11 (computationally) = 0.230561023

K11 (by hand) = 0.230560779

x2= 0.159481985

x3= 0.176675955

x1= 0.320697341

T by hand=509.212529

V by hand=−150.382206

```
m2=26.
Energy without mass = 651.196578
Energy=  16747.1966
R=  0.0250861775
normx2 (computationally)=  -0.62639146
normx2 (by hand)=  -0.62639153
normx3 (computationally)=  -0.582341804
normx3 (by hand)=  -0.582341835
normx1 (computationally)=  -0.41806271
normx1 (by hand)=  -0.418062719
0 to x2 (h4 computationally)=  4.14665127
0 to x2 (h4 by hand)=  4.14665126
x2 to x3 (h5 computationally)=  0.826367637
x2 to x3 (h5 by hand)=  0.826368113
x2 to x3 (h5 by handdddd)=  0.82636791
x3 to x1 (h6 computationally)=  1.83834682
x3 to x1 (h6 by hand)=  1.83834662
K22  (computationally) = 0.210915855
K22  (by hand) = 0.210915851
K32  (computationally) = 0.199448452
K32  (by hand) = 0.199448449
K21  (computationally) = 0.179191092
K21  (by hand) =0.179191092
K31  (computationally) = 0.177948951
K31  (by hand) = 0.177949044
K33  (computationally) = 0.190696575
K33  (by hand) = 0.190696875
K11  (computationally) = 0.231032201
K11  (by hand) = 0.231031936
x2= 0.15916842
x3= 0.176326553
x1=  0.320033896
T by hand=511.304012
V by hand=-150.685124
m2=26.1
Energy without mass = 651.174389
```

```
Energy=  16747.1744
R=  0.0250861775
normx2 (computationally)=  -0.626391471
normx2 (by hand)=  -0.62639153
normx3 (computationally)=  -0.582341803
normx3 (by hand)=  -0.582341835
normx1 (computationally)=  -0.418062707
normx1 (by hand)=  -0.418062719
0 to x2 (h4 computationally)=  4.16293236
0 to x2 (h4 by hand)=  4.16293238
x2 to x3 (h5 computationally)=  0.829685547
x2 to x3 (h5 by hand)=  0.829685956
x2 to x3 (h5 by handdddd)=  0.829685823
x3 to x1 (h6 computationally)=  1.84600668
x3 to x1 (h6 by hand)=  1.84600645
K22  (computationally) = 0.211326693
K22  (by hand) = 0.2113267
K32  (computationally) = 0.199837793
K32  (by hand) = 0.199837798
K21  (computationally) = 0.179551536
K21  (by hand) =0.179551537
K31  (computationally) = 0.17830705
K31  (by hand) = 0.17830713
K33  (computationally) = 0.191069453
K33  (by hand) = 0.191069716
K11  (computationally) = 0.231502451
K11  (by hand) = 0.231502154
x2= 0.158856682
x3= 0.175979198
x1=  0.319374467
T by hand=513.395739
V by hand=-150.987465
m2=26.2
Energy without mass = 651.171407
Energy=  16747.1714
R=  0.0250861775
```

```
normx2 (computationally)=  -0.6263914
normx2 (by hand)=  -0.62639153
normx3 (computationally)=  -0.582341803
normx3 (by hand)=  -0.582341835
normx1 (computationally)=  -0.41806271
normx1 (by hand)=  -0.418062719
0 to x2 (h4 computationally)=  4.17921365
0 to x2 (h4 by hand)=  4.1792134
x2 to x3 (h5 computationally)=  0.833004584
x2 to x3 (h5 by hand)=  0.833005436
x2 to x3 (h5 by handdddd)=  0.833004861
x3 to x1 (h6 computationally)=  1.8536677
x3 to x1 (h6 by hand)=  1.8536675
K22  (computationally) = 0.211736583
K22  (by hand) = 0.211736518
K32  (computationally) = 0.20022631
K32  (by hand) = 0.200226261
K21  (computationally) = 0.179911285
K21  (by hand) =0.179911278
K31  (computationally) = 0.178664602
K31  (by hand) = 0.178664763
K33  (computationally) = 0.191442043
K33  (by hand) = 0.191442546
K11  (computationally) = 0.231971919
K11  (by hand) = 0.231971671
x2= 0.158546748
x3= 0.175633879
x1=  0.31871902
T by hand=515.487756
V by hand=-151.28933
m2=26.3
Energy without mass = 651.187335
Energy=  16747.1873
R=  0.0250861775
normx2 (computationally)=  -0.62639142
normx2 (by hand)=  -0.62639153
```

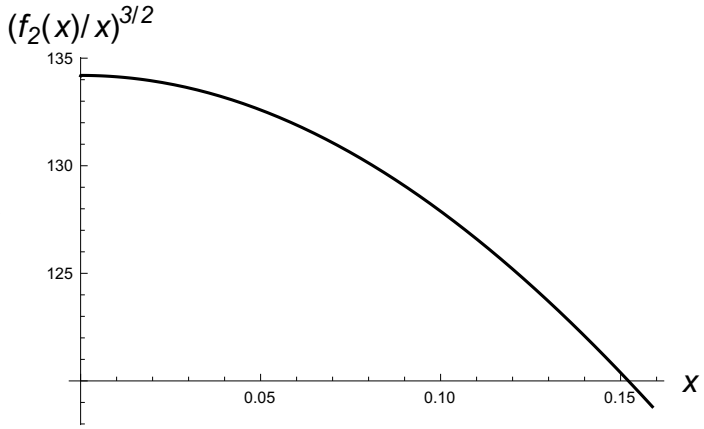
```
normx3 (computationally)=  -0.582341803
normx3 (by hand)=  -0.582341835
normx1 (computationally)=  -0.418062717
normx1 (by hand)=  -0.418062719
0 to x2 (h4 computationally)=  4.16456017
0 to x2 (h4 by hand)=  4.16456
x2 to x3 (h5 computationally)=  0.830017892
x2 to x3 (h5 by hand)=  0.83001862
x2 to x3 (h5 by handdddd)=  0.830018168
x3 to x1 (h6 computationally)=  1.84677284
x3 to x1 (h6 by hand)=  1.84677268
K22  (computationally) = 0.211367629
K22  (by hand) = 0.211367584
K32  (computationally) = 0.199876635
K32  (by hand) = 0.199876601
K21  (computationally) = 0.179587546
K21  (by hand) =0.179587541
K31  (computationally) = 0.178342916
K31  (by hand) = 0.178343055
K33  (computationally) = 0.191106972
K33  (by hand) = 0.191107407
K11  (computationally) = 0.231549614
K11  (by hand) = 0.231549432
x2= 0.158825604
x3= 0.175944581
x1=  0.319308751
T by hand=513.604978
V by hand=-151.017773
m2=26.21
Energy without mass = 651.172125
Energy=  16747.1721
R=  0.0250861775
normx2 (computationally)=  -0.626391442
normx2 (by hand)=  -0.62639153
normx3 (computationally)=  -0.582341803
normx3 (by hand)=  -0.582341835
```

```
normx1 (computationally)=  -0.418062708
normx1 (by hand)=  -0.418062719
0 to x2 (h4 computationally)=  4.16618846
0 to x2 (h4 by hand)=  4.16618837
x2 to x3 (h5 computationally)=  0.830349489
x2 to x3 (h5 by hand)=  0.830350081
x2 to x3 (h5 by handdddd)=  0.830349765
x3 to x1 (h6 computationally)=  1.8475388
x3 to x1 (h6 by hand)=  1.84753858
K22  (computationally) = 0.211408703
K22  (by hand) = 0.211408681
K32  (computationally) = 0.19991554
K32  (by hand) = 0.199915524
K21  (computationally) = 0.17962354
K21  (by hand) =0.179623538
K31  (computationally) = 0.178378641
K31  (by hand) = 0.178378755
K33  (computationally) = 0.191144106
K33  (by hand) = 0.191144469
K11  (computationally) = 0.231596432
K11  (by hand) = 0.231596156
x2= 0.15879455
x3= 0.175909975
x1=  0.319243062
T by hand=513.81413
V by hand=-151.047901
m2=26.22
Energy without mass = 651.173078
Energy=  16747.1731
R=  0.0250861775
normx2 (computationally)=  -0.626391422
normx2 (by hand)=  -0.62639153
normx3 (computationally)=  -0.582341803
normx3 (by hand)=  -0.582341835
normx1 (computationally)=  -0.418062714
normx1 (by hand)=  -0.418062719
```

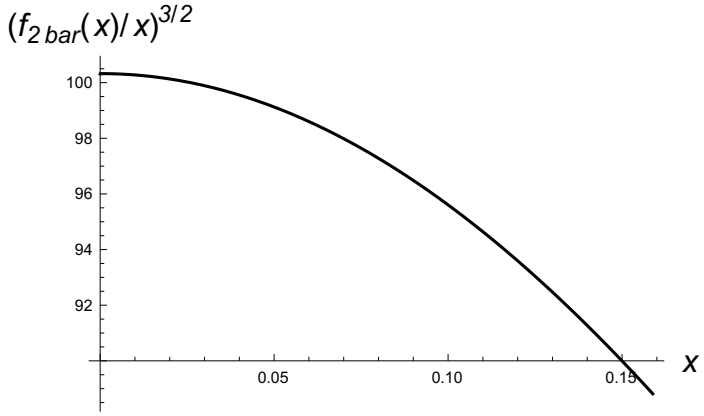
```
0 to x2 (h4 computationally)=  4.164723
0 to x2 (h4 by hand)=  4.16472284
x2 to x3 (h5 computationally)=  0.83005105
x2 to x3 (h5 by hand)=  0.830051762
x2 to x3 (h5 by handdddd)=  0.830051325
x3 to x1 (h6 computationally)=  1.84684941
x3 to x1 (h6 by hand)=  1.84684924
K22  (computationally) = 0.211371737
K22  (by hand) = 0.211371695
K32  (computationally) = 0.199880526
K32  (by hand) = 0.199880494
K21  (computationally) = 0.179591144
K21  (by hand) =0.17959114
K31  (computationally) = 0.178346488
K31  (by hand) = 0.178346623
K33  (computationally) = 0.191110684
K33  (by hand) = 0.191111112
K11  (computationally) = 0.231554268
K11  (by hand) = 0.23155406
x2= 0.158822498
x3= 0.175941119
x1=  0.319302179
T by hand=513.625887
V by hand=-151.020776
m2=26.211
Energy without mass = 651.172213
Energy=  16747.1722
R=  0.0250861775
normx2 (computationally)=  -0.626391425
normx2 (by hand)=  -0.62639153
normx3 (computationally)=  -0.582341803
normx3 (by hand)=  -0.582341835
normx1 (computationally)=  -0.418062719
normx1 (by hand)=  -0.418062719
0 to x2 (h4 computationally)=  4.16488583
0 to x2 (h4 by hand)=  4.16488568
```

```
x2 to x3 (h5 computationally)=  0.830084207
x2 to x3 (h5 by hand)=  0.830084904
x2 to x3 (h5 by handdddd)=  0.830084482
x3 to x1 (h6 computationally)=  1.84692607
x3 to x1 (h6 by hand)=  1.84692593
K22  (computationally) = 0.211375846
K22  (by hand) = 0.211375806
K32  (computationally) = 0.199884417
K32  (by hand) = 0.199884387
K21  (computationally) = 0.179594748
K21  (by hand) =0.179594744
K31  (computationally) = 0.178350064
K31  (by hand) = 0.178350197
K33  (computationally) = 0.191114397
K33  (by hand) = 0.191114816
K11  (computationally) = 0.231559041
K11  (by hand) = 0.231558879
x2= 0.158819392
x3= 0.175937658
x1=  0.319295612
T by hand=513.646821
V by hand=-151.023821
m2=26.212
Energy without mass = 651.172297
Energy=  16747.1723
x1=0.319295612
x3=0.175937658
x2=0.158819392
Kinetic Energy=513.646821
Potential Energy=-151.023821
Volume Energy=288.549297
```

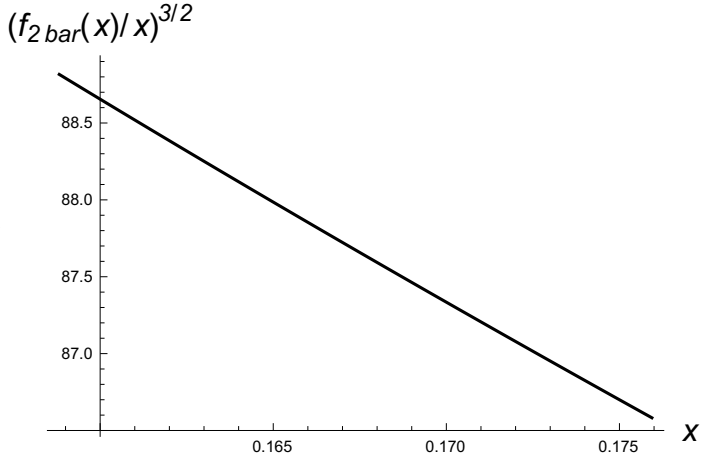
Out[ ]=


$(f_2(x)/x)^{3/2}$
135
130
125
0.05
0.10
0.15
x

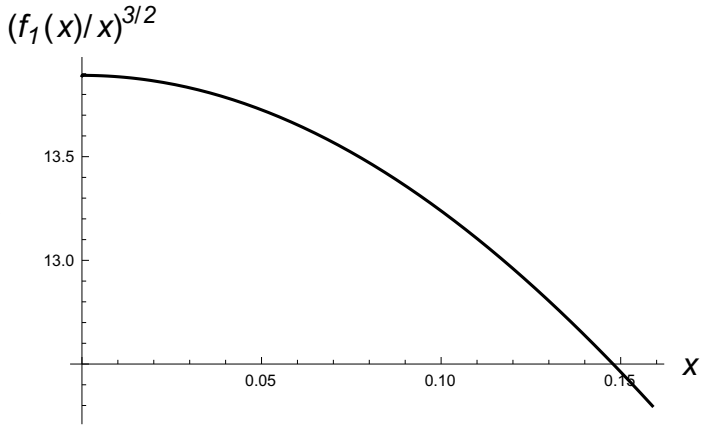
Out[ ]=


$(f_{2\,bar}(x)/x)^{3/2}$
100
98
96
94
92
0.05
0.10
0.15
x

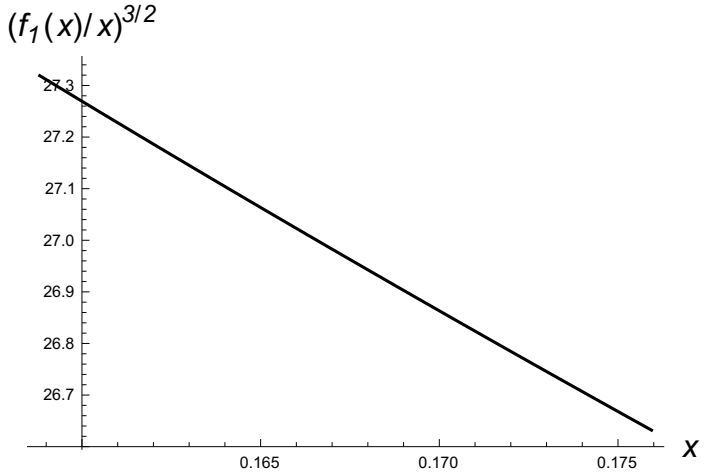
Out[ ]=


$(f_{2\,bar}(x)/x)^{3/2}$
88.5
88.0
87.5
87.0
0.165
0.170
0.175
x

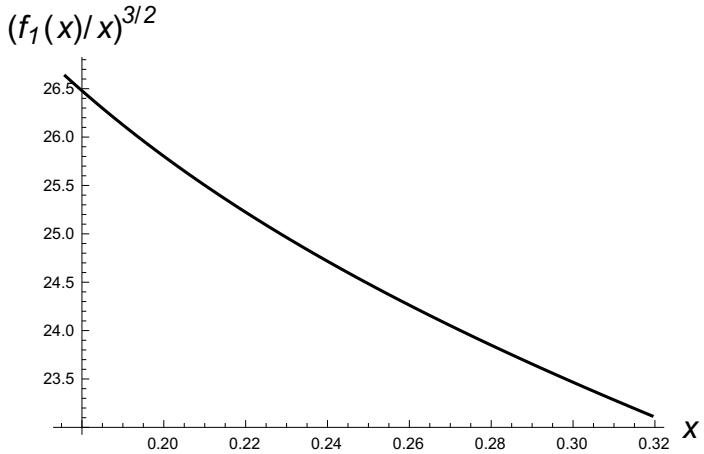
Out[ ]=


$(f_1(x)/x)^{3/2}$
13.5
13.0
0.05
0.10
0.15
x

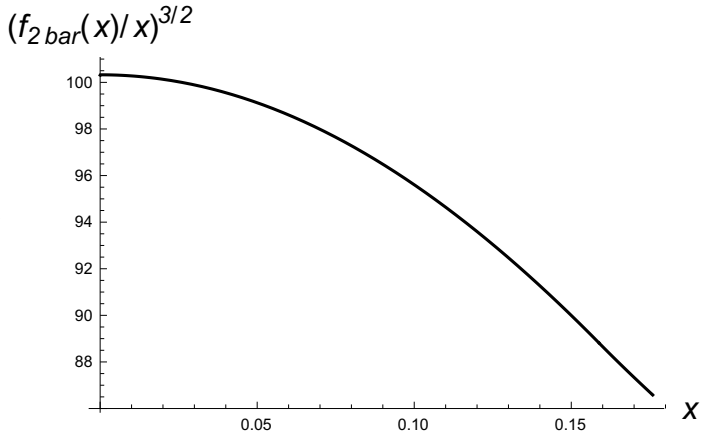
Out[ ]=


$(f_1(x)/x)^{3/2}$
27.3
27.2
27.1
27.0
26.9
26.8
26.7
0.165
0.170
0.175
x

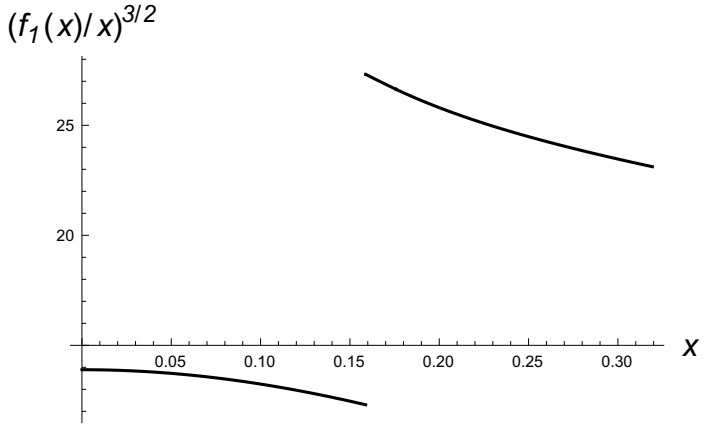
Out[ ]=


$(f_1(x)/x)^{3/2}$
26.5
26.0
25.5
25.0
24.5
24.0
23.5
0.20
0.22
0.24
0.26
0.28
0.30
0.32
x

Out[ ]=



Out[ ]=



General: Further output of FindMinimum::fmdig will be suppressed during this calculation.

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

General: Further output of FindMinimum::lstol will be suppressed during this calculation.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

General: Further output of FindMinimum::fmdig will be suppressed during this calculation.

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

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General: Further output of FindMinimum::lstol will be suppressed during this calculation.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

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